

Source 01: Dashboard Use Case Evidence

Source-summary notes — existing heat-risk dashboards and alerting systems reviewed for the Heat Risk Dashboard project.

Problem Gap

Heat-health alerting systems (NWS/CDC HeatRisk, UKHSA Heat-Health Alerts, national systems backed by WHO/WMO) reliably issue risk information, but evidence shows this does not reliably become coordinated action. A 2026 UK study found 91% of people intended to change behavior during a heat alert, but only 41.6% actually took protective action (ScienceDirect, 2026). Governance research in Western Cape, South Africa found health institutions are often limited to reactive alerting rather than embedded preparedness or cross-sector response (medRxiv, 2025).

Users and Decisions

Public health officers need to know which zones are at risk and where coverage gaps exist. Community health workers need a prioritized list of who to check on. Facilities (clinics, shelters, cooling centers) need to know if an alert changes their protocol and need to report capacity/status. Residents and caregivers need to know personal risk level and whether someone has been assigned to check on them.

Dashboard Capabilities Observed in Existing Tools

- NWS HeatRisk Forecast: 7-day, 5-level color-numeric risk index by location, built for state/local health officials.
- CDC HeatRisk Dashboard: consumer-facing zip-code lookup with personalized guidance and resource links.
- CDC Heat & Health Tracker: county-level ED visit rates from NSSP, historical heat-related illness burden, vulnerability/ZCTA data.
- Harris County (Houston) Heat and Health Dashboard: 4-page dashboard with an interactive ZCTA map layering illness data, social vulnerability, and built-environment factors, updated daily with a one-day data lag.
- HHS/NEMSIS heat-EMS dashboard: state/county heat-related EMS activation counts from 911 data, broken down by demographics.
- UKHSA Weather Health Alerting System: region-by-region alert status board tied to an Adverse Weather and Health Plan.

Implementation Constraints and Evidence Gaps

Every reviewed tool stops at displaying current risk level or historical burden. None of them include a task, assignment, or completion-tracking layer. Inconsistent local data collection also limits the ability to translate a weather forecast into a measurable, targetable human-health picture (Kansas Reflector, 2026). This is the specific gap the proposed dashboard is meant to close: making the operational response to an alert visible and trackable, not just the alert itself.

Full source list and citations are provided in Belle_literature_review.md.